Clinical Dataset Anomaly Detector (Medical AI Auditing)

This tool provides an automated pipeline to audit clinical and medical datasets for structural anomalies, data fabrication signatures, and distribution shifts. Inspired by recent investigations into unverified, simulated, or "dubious" datasets used to train public machine learning models for diseases like stroke and diabetes, this tool helps researchers ensure data authenticity before clinical deployment.

■ The Problem

In medical data science, many public open-source datasets contain synthetic mathematical patterns, lack proper clinical provenance, or exhibit structural anomalies indicating fabrication. Models trained on such data learn artificial correlations rather than true biological signals, rendering them dangerous if deployed on real patients.

■■ Core Analytical Features

1. **Missing Data Completeness Check:** Real-world electronic health records (EHR) naturally contain missing entries due to fragmented clinical workflows. A dataset with exactly 0% missing fields across thousands of entries is flagged as highly suspicious for synthetic generation.
2. **Sequential Distribution Shift Detection:** Detects abrupt mathematical jumps or structural drops in clinical variables (e.g., Blood Glucose, Blood Pressure) relative to patient IDs. This identifies cases where distinct synthetic batches or unaligned data pools were stitched together.
3. **Oversampling & Duplicate Check:** Identifies exact row duplication often caused by naive implementation of data-augmentation algorithms (like SMOTE) to unnaturally bloat clinical datasets.

■ Getting Started

Ensure you have Python 3.8+ and the core dependencies installed:

```
pip install pandas numpy matplotlib
```

■ Usage Example

```
from generator import generate_medical_dataset
from detector import analyze_and_detect_anomalies

# 1. Generates a dataset with intentional structural anomalies
dubious_df = generate_medical_dataset(num_patients=1000, make_too_perfect=True,
inject_distribution_shift=True)
dubious_df.to_csv('synthetic_medical_data.csv', index=False)

# 2. Run diagnostics on the generated file
analyze_and_detect_anomalies('synthetic_medical_data.csv')
```

■ Sample Diagnostic Output Logs

```
--- Loading Dataset ---
Dataset Shape: 1000 rows, 7 columns

--- 1. Missing Data Check ---
[■■ WARNING] 0% missing values detected. Real-world routine clinical data
almost always contains missing fields. This could indicate a synthetic dataset.

--- 2. Patient ID vs. Feature Distribution Shift Check ---
[■ CRITICAL] Structural shift detected in 'Avg_Glucose_Level' based on ID tracking!
First half mean: 104.22, Second half mean: 171.45

--- Conclusion ---
■ Summary: This dataset exhibits structural anomalies typical of unverified,
simulated, or improperly audited clinical data. Do not deploy models built on this data.
```

■ Scientific Rigor & Best Practices

Passing mathematical checks does not automatically mean a dataset is clinically valid. When reviewing data for medical AI, always audit the **Data Provenance**:

- **Who** collected the data? (Affiliated hospital, clinical trial network)
- **When** and **How** were the measurements taken? (Laboratory assay criteria, machine specs)
- Are the distributions aligned with established clinical literature thresholds?
- Consult reporting standards like **TRIPOD+AI** (Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis + Artificial Intelligence) for comprehensive clinical validation.